

Divergence Theorem

Lecture 11

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Divergence Theorem

Inquiry Question:

- Idea of Flux in 3—dimension
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- The **divergence form of Green's Theorem** in the plane states that the net outward flux of a vector field across a simple closed curve can be calculated by integrating the divergence of the field over the region enclosed by the curve.
 - The corresponding theorem in three dimensions, called the **Divergence Theorem**, states that the net outward flux of a vector field across a closed surface in space can be calculated by integrating the divergence of the field over the region enclosed by the surface.
 - In this section, we prove the **Divergence Theorem** and show how it simplifies the calculation of flux, which is the integral of vector field over the closed oriented surface.
 - We also derive Gauss's law for flux in an electric field and the continuity equation of hydrodynamics.

Divergence

The divergence of a vector field $\mathbf{F} = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$ is the scalar function

$$\operatorname{div} \mathbf{F} = \vec{\nabla} \cdot \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}$$

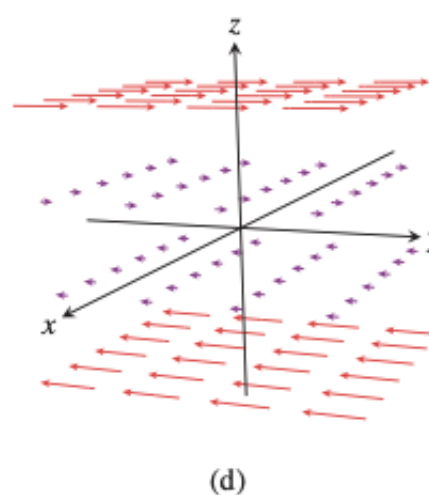
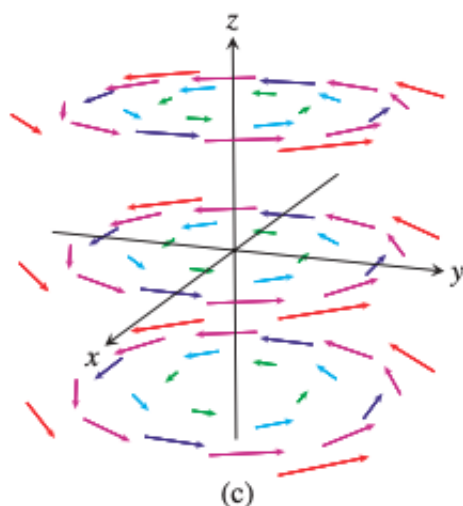
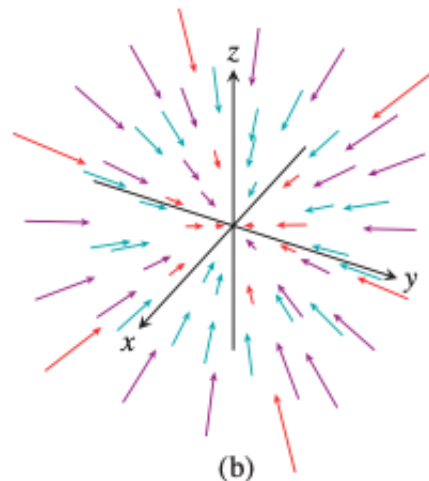
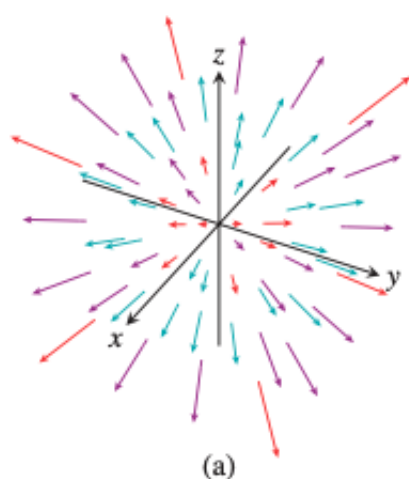
The symbol “ $\operatorname{div} \mathbf{F}$ ” is read as “divergence of $\mathbf{F}$ ” or “ $\operatorname{div} \mathbf{F}$.” The notation $\vec{\nabla} \cdot \mathbf{F}$ is read as “del dot $\mathbf{F}$.”

$\operatorname{div} \mathbf{F}$ has the same physical interpretation in three dimensions as it does in two. **If $\mathbf{F}$ is the velocity field of a flowing gas, the value of $\operatorname{div} \mathbf{F}$ at a point (x, y, z) is the rate at which the gas is compressing or expanding at (x, y, z) .**

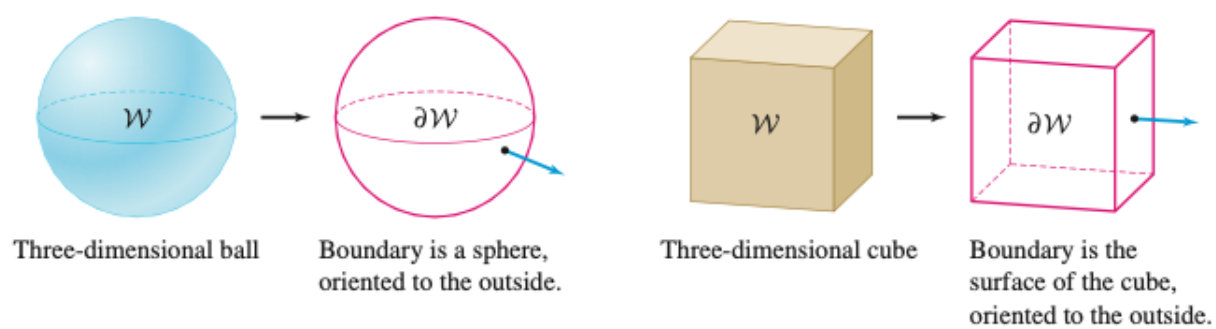
The divergence is the flux per unit volume or flux density at the point.

The following vector fields represent the velocity of a gas flowing in space. Find the divergence of each vector field and interpret its physical meaning.

- (a) **Expansion:** $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$
- (b) **Compression:** $\mathbf{F}(x, y, z) = -x\mathbf{i} - y\mathbf{j} - z\mathbf{k}$
- (c) **Rotation about the z -axis:** $\mathbf{F}(x, y, z) = -y\mathbf{i} + x\mathbf{j}$
- (d) **Shearing along parallel horizontal planes:**
 $\mathbf{F}(x, y, z) = z\mathbf{j}$



1. A **closed surface** is a surface that completely encloses a volume and has no edges or boundaries. It has a distinct "inside" and "outside," and you cannot travel from one to the other without crossing the surface. Examples include the surface of a sphere, a cube, or a cylinder with its top and bottom ends capped.
- $\mathcal{S}$ is a **closed surface** that encloses a three-dimensional (or 3D) region $\mathcal{W}$



- We consider a piecewise smooth closed surface $\mathcal{S}$, which means $\mathcal{S}$ consists of one smooth surface or at most finitely many smooth surfaces that have been glued together along their boundaries, as in the example of the cube.

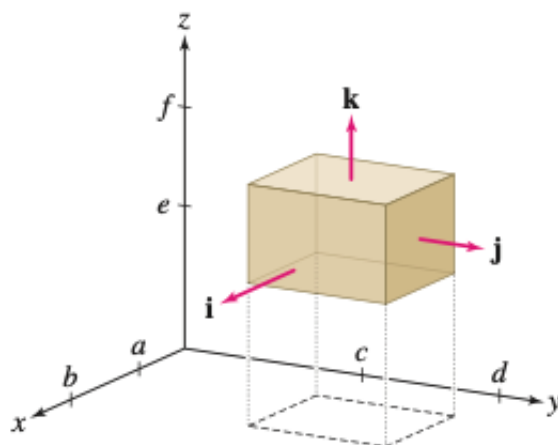
Divergence Theorem

- Let $\mathcal{S}$ be a closed surface that encloses a region $\mathcal{W}$ in $\mathbb{R}^3$.
- Assume that $\mathcal{S}$ is piecewise smooth and is oriented by normal vectors pointing to the outside of $\mathcal{W}$.
- Let $\mathbf{F}$ be a vector field whose domain contains $\mathcal{W}$.

Then

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_{\mathcal{W}} \vec{\nabla} \cdot \mathbf{F} \, dV$$

We prove the Divergence Theorem in the special case that $\mathcal{W}$ is a box $[a, b] \times [c, d] \times [e, f]$ as in Figure below.



The proof can be modified to treat more general regions such as the interiors of spheres and cylinders.

We write each side in the manner

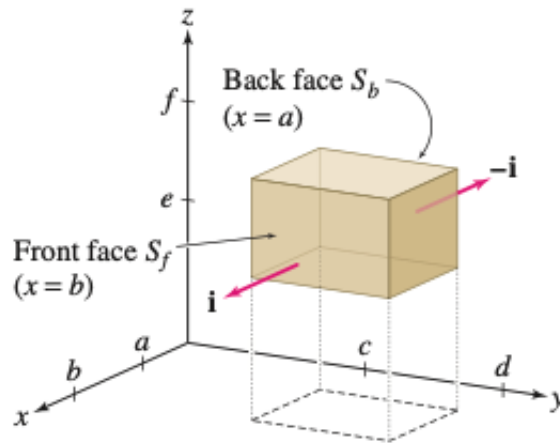
$$\begin{aligned} \iint_S (F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}) \cdot \mathbf{n} \, d\sigma &= \iint_S F_1 \mathbf{i} \cdot \mathbf{n} \, d\sigma \\ &+ \iint_S F_2 \mathbf{j} \cdot \mathbf{n} \, d\sigma \\ &+ \iint_S F_3 \mathbf{k} \cdot \mathbf{n} \, d\sigma \end{aligned}$$

This transforms into

$$\begin{aligned} \iiint_W \operatorname{div} (F_1 \mathbf{i} + F_2 \mathbf{j} + F_3 \mathbf{k}) dV &= \iiint_{\mathcal{W}} \operatorname{div}(F_1 \mathbf{i}) dV \\ &+ \iiint_{\mathcal{W}} \operatorname{div}(F_2 \mathbf{j}) dV \\ &+ \iiint_{\mathcal{W}} \operatorname{div}(F_3 \mathbf{k}) dV \end{aligned}$$

As in the proofs of Green's and Stokes' Theorems, we show that corresponding terms are equal. It will suffice to carry out the argument for the i -component (the other two components are similar). Thus, we assume that $\mathbf{F} = F_1 \mathbf{i}$.

The surface integral over boundary $\mathcal{S}$ of the box is the sum of the integrals over the six faces. However, $\mathbf{F} = F_1 \mathbf{i}$ is orthogonal to the normal vectors to the top and bottom as well as the two side faces because $\mathbf{F} \cdot \mathbf{j} = \mathbf{F} \cdot \mathbf{k} = 0$. Therefore, the surface integrals over these faces are zero. Nonzero contributions come only from the front and back faces, which we denote S_f and S_b and evident from the figure below:



This implies that

$$\iint_S \mathbf{F} \cdot \mathbf{n} d\sigma = \iint_{S_f} \mathbf{F} \cdot \mathbf{n} d\sigma + \iint_{S_b} \mathbf{F} \cdot \mathbf{n} d\sigma$$

To evaluate these integrals, we parametrize S_f and S_b by

$$\begin{aligned} G_f(y, z) &= (b, y, z), \quad c \leq y \leq d, \quad e \leq z \leq f \\ G_b(y, z) &= (a, y, z), \quad c \leq y \leq d, \quad e \leq z \leq f \end{aligned}$$

The normal vectors for these parametrizations are

$$\begin{aligned} \frac{\partial G_f}{\partial y} \times \frac{\partial G_f}{\partial z} &= \mathbf{j} \times \mathbf{k} = \mathbf{i} \\ \frac{\partial G_b}{\partial y} \times \frac{\partial G_b}{\partial z} &= \mathbf{j} \times \mathbf{k} = \mathbf{i} \end{aligned}$$

However, the outward-pointing normal for S_b is $-\mathbf{i}$, so a minus sign is indeed in the surface integral over S_b using the parametrization G_b :

$$\begin{aligned} \iint_{S_f} \mathbf{F} \cdot \mathbf{n} \, d\sigma + \iint_{S_b} \mathbf{F} \cdot \mathbf{n} \, d\sigma &= \int_e^f \int_c^d F_1(b, y, z) dy dz \\ &\quad - \int_e^f \int_c^d F_1(a, y, z) dy dz \\ &= \int_e^f \int_c^d (F_1(b, y, z) - F_1(a, y, z)) dy dz \end{aligned}$$

By using Fundamental Theorem of Calculus in One Variable

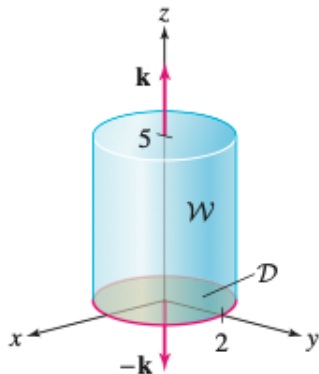
$$F_1(b, y, z) - F_1(a, y, z) = \int_a^b \frac{\partial F_1}{\partial x}(x, y, z) dx$$

Since $\operatorname{div}(\mathbf{F}) = \operatorname{div}(F_1 \mathbf{i}) = \frac{\partial F_1}{\partial x}$, we obtain the desired result

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma &= \int_e^f \int_c^d \int_a^b \frac{\partial F_1}{\partial x}(x, y, z) dx dy dz \\ &= \iiint_{\mathcal{W}} \operatorname{div}(\mathbf{F}) dV \end{aligned}$$

Example

Verify Divergence Theorem for $\mathbf{F}(x, y, z) = \langle y, yz, z^2 \rangle$ and the cylinder given below:



- Integrate over the side of the cylinder. This can be achieved via parametrization

$$\vec{r}(u, v) = \langle 2 \cos(\theta), 2 \sin(\theta), z \rangle$$

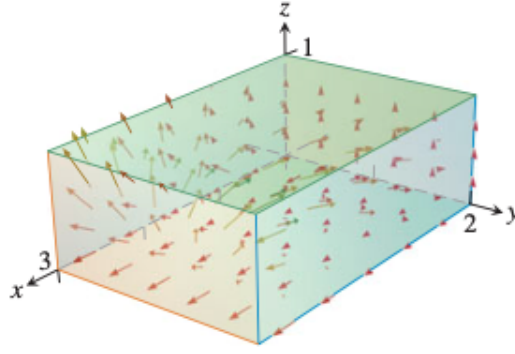
where $0 \leq \theta \leq 2\pi$ and $0 \leq z \leq 5$.

- Integrate over the top and the bottom of the cylinder.
- Find the total flux (sides+top+bottom).
- Compare with the integral of divergence (Divergence Theorem)

Hint: Answer: 150π

Example

- Calculate the flux of the vector field $\mathbf{F}(x, y, z) = \langle x^2, 4xyz, ze^x \rangle$ out of the box-shaped region $D : 0 \leq x \leq 3, 0 \leq y \leq 2, 0 \leq z \leq 1$
- Integrate $\text{div } \mathbf{F}$ over this region and show that the result is the same value as in part (a), as asserted by the Divergence Theorem.



The region D has six sides. We calculate the flux across each side in turn. Consider the top side in the plane $z = 1$, having outward normal $\mathbf{n} = \mathbf{k}$. The flux across this side is given by $\mathbf{F} \cdot \mathbf{n} = ze^x$. Since, $z = 1$ on this side, the flux at the point (x, y, z) on the top is e^x . The outward flux across this side is given by the surface integral

$$\int_0^2 \int_0^3 e^x dx dy = 2e^3 - 2$$

The outward flux across the other sides is computed similarly, and the results are summarized in the table below:

Side	Unit normal n	$\mathbf{F} \cdot \mathbf{n}$	Flux across side
$x = 0$	$-\mathbf{i}$	$-x^2 = 0$	0
$x = 3$	$\mathbf{i}$	$x^2 = 9$	18
$y = 0$	$-\mathbf{j}$	$-4xyz = 0$	0
$y = 2$	$\mathbf{j}$	$4xyz = 8xz$	18
$z = 0$	$-\mathbf{k}$	$-ze^x = 0$	0
$z = 1$	$\mathbf{k}$	$ze^x = e^x$	$2e^3 - 2$

The total flux is obtained by adding flux across each side

$$\text{Total Flux} = 18 + 18 + 2e^3 - 2 = 34 + 2e^3$$

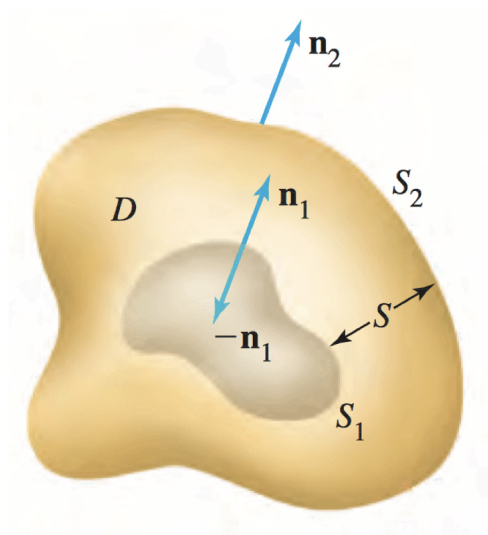
Part (b): Verify using Divergence Theorem

Divergence Theorem

Inquiry Question: Divergence Theorem for regions bounded by two surfaces

The Divergence Theorem may be extended to more general solid regions.

1. We consider the important case of hollow regions.
2. Suppose D is a region consisting of all points inside a closed oriented surface S_2 and outside a closed oriented surface S_1 , where S_1 lies within S_2 .
3. The boundary of D consists of S_1 and S_2 .
4. Note that D is simply connected.



5. We let $\vec{n}_1$ and $\vec{n}_2$ be the outward unit normal vectors for S_1 and S_2 , respectively. Note that $\vec{n}_1$ points into D , so the outward normal to S on S_1 is $-\vec{n}_1$. With this observation, the Divergence Theorem takes the following form.

Divergence Theorem for region between two surfaces

Suppose the vector field $\vec{F}$ satisfies the conditions of the Divergence Theorem on a region D bounded by two smooth oriented surfaces S_1 and S_2 , where S_1 lies within S_2 . Let S be the entire boundary of D (i.e., $S = S_1 \cup S_2$), and let $\vec{n}_1$ and $\vec{n}_2$ be the outward unit normal vectors for S_1 and S_2 , respectively. Then

$$\iiint_D (\vec{\nabla} \cdot \vec{F}) dV = \iint_S \vec{F} \cdot \vec{n} d\sigma = \iint_{S_2} \vec{F} \cdot \vec{n}_2 d\sigma - \iint_{S_1} \vec{F} \cdot \vec{n}_1 d\sigma.$$

Remark: This form of the Divergence Theorem is applicable to vector fields that are not differentiable at the origin, as is the case with some important radial vector fields.

Example

Consider the inverse square vector field

$$\vec{F} = \frac{\vec{r}}{|\vec{r}|^3} = \frac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{3/2}}.$$

- (a) Find the net outward flux of $\vec{F}$ across the surface of the region

$$D = \{(x, y, z) : a^2 \leq x^2 + y^2 + z^2 \leq b^2\}$$

that lies between two concentric spheres with radii a and b .

- (b) Find the outward flux of $\vec{F}$ across any sphere that encloses the origin.

Consider the inverse square vector field

$$\vec{F} = \frac{\vec{r}}{|\vec{r}|^3} = \frac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{3/2}}.$$

Although this **field is undefined** at the **origin**, it is well-defined and differentiable everywhere else. We will compute the net out-

ward flux of $\vec{F}$ across a closed surface that encloses the origin, using the Divergence Theorem.

Let D be the region between two concentric spheres centered at the origin with radii a and b :

$$D = \{(x, y, z) : a^2 \leq x^2 + y^2 + z^2 \leq b^2\}.$$

Its boundary consists of two surfaces:

$$S = S_1 \cup S_2,$$

where S_1 is the inner sphere of radius a , and S_2 is the outer sphere of radius b .

From vector calculus, the divergence of the general radial field

$$\vec{F} = \frac{\vec{r}}{|\vec{r}|^p}$$

is given by

$$\vec{\nabla} \cdot \vec{F} = \frac{3 - p}{|\vec{r}|^p}.$$

For $p = 3$, we get

$$\vec{\nabla} \cdot \vec{F} = 0$$

for all $\vec{r} \neq \vec{0}$.

Because the origin is excluded from D , $\vec{F}$ is differentiable throughout D , and therefore

$$\iiint_D (\nabla \cdot \vec{F}) dV = 0.$$

By the Divergence Theorem,

$$\iint_S \vec{F} \cdot \vec{n} d\sigma = \iiint_D (\nabla \cdot \vec{F}) dV = 0.$$

Since S is composed of S_1 and S_2 , we can write

$$\iint_S \vec{F} \cdot \vec{n} d\sigma = \iint_{S_2} \vec{F} \cdot \vec{n}_2 d\sigma - \iint_{S_1} \vec{F} \cdot \vec{n}_1 d\sigma = 0,$$

where the negative sign arises because the outward normal of S_1 (with respect to D) points inward toward the origin. Hence, the net flux through S is zero.

This implies that

$$\iint_{S_2} \vec{F} \cdot \vec{n}_2 d\sigma = \iint_{S_1} \vec{F} \cdot \vec{n}_1 d\sigma.$$

That is, the flux out of D across S_2 equals the flux into D across S_1 .

Flux Across the Inner Sphere:

To compute the flux, consider the smaller sphere S_1 by considering $\vec{n} = \frac{\vec{\nabla}g}{|\vec{\nabla}g|}$, where $g(x, y, z) = x^2 + y^2 + z^2 - a^2$, and $|\vec{r}| = a$. The outward unit normal vector on S_1 is

$$\vec{n}_1 = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{r}}{a}.$$

Hence, the surface integral becomes

$$\iint_{S_1} \vec{F} \cdot \vec{n}_1 d\sigma = \iint_{S_1} \frac{\vec{r}}{|\vec{r}|^3} \cdot \frac{\vec{r}}{|\vec{r}|} d\sigma = \iint_{S_1} \frac{\vec{r} \cdot \vec{r}}{|\vec{r}|^4} d\sigma.$$

Since $\vec{r} \cdot \vec{r} = |\vec{r}|^2 = a^2$, we have

$$\iint_{S_1} \vec{F} \cdot \vec{n}_1 d\sigma = \iint_{S_1} \frac{a^2}{a^4} d\sigma = \iint_{S_1} \frac{1}{a^2} d\sigma.$$

The surface area of S_1 is $4\pi a^2$, so

$$\iint_{S_1} \vec{F} \cdot \vec{n}_1 d\sigma = \frac{1}{a^2}(4\pi a^2) = 4\pi.$$

Conclusion

Therefore, the outward flux of $\vec{F}$ across S_1 is

$$\boxed{\Phi = 4\pi.}$$

Because the flux is the same across any sphere enclosing the origin, the result holds for S_2 or any smooth closed surface enclosing the origin.

Interpretation: The inverse-square field $\vec{F} = \frac{\vec{r}}{|\vec{r}|^3}$ is unique among radial fields in that its total outward flux through any closed surface enclosing the origin is constant (4π), independent of the radius of the surface.

Gauss's Law: An Application of Divergence Theorem

Applying the Divergence Theorem to electric fields leads to one of the fundamental results in physics — **Gauss's Law**.

Interpretation: It states that the flux of the electric field out of an arbitrary closed surface is proportional to the electric charge enclosed by the surface, irrespective of how that charge is distributed.

The electric field due to a point charge Q located at the origin is given by the inverse square law:

$$\vec{E}(x, y, z) = \frac{Q}{4\pi\epsilon_0} \frac{\vec{r}}{|\vec{r}|^3},$$

where $\vec{r} = \langle x, y, z \rangle$ and ϵ_0 is a physical constant called the *permittivity of free space*.

From the calculation in the previous Example, the flux of the field

$$\frac{\vec{r}}{|\vec{r}|^3}$$

across any closed surface that encloses the origin is 4π . Therefore, the total flux of the electric field $\vec{E}$ across any such surface is

$$\Phi_E = \iint_S \vec{E} \cdot \vec{n} d\sigma = \frac{Q}{4\pi\epsilon_0} \cdot 4\pi = \frac{Q}{\epsilon_0}.$$

This leads directly to the integral form of **Gauss's Law**:

$$\boxed{\iint_S \vec{E} \cdot \vec{n} d\sigma = \frac{Q_{\text{enc}}}{\epsilon_0}}$$

where Q_{enc} is the total electric charge enclosed by the closed surface S .

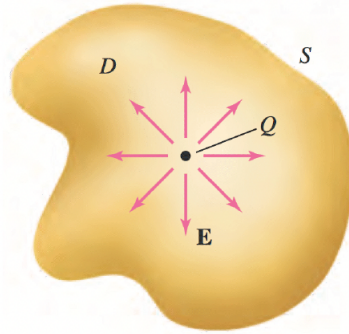


Figure 1: **Gauss's Law:** Flux of electric field across S due to a point charge Q $\iint_S \vec{E} \cdot \vec{n} d\sigma = \frac{Q}{\epsilon_0}$

Physically, Gauss's Law states that the total electric flux through a closed surface is proportional to the total charge enclosed within that surface.

Also,

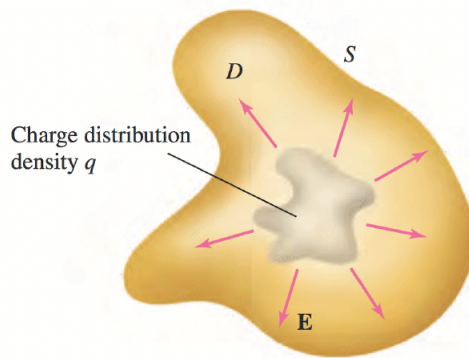


Figure 2: **Gauss's Law:** Flux of electric field across S due to charge distribution q $\iint_S \vec{E} \cdot \vec{n} d\sigma = \frac{1}{\epsilon_0} \iiint_D q dV$